AAMAS 2027 — Blue Sky Ideas Track

Additional experiments

Learning Signed Critic Errors: A Probabilistic Approach to Addressing Function Approximation Errors in Actor–Critic Reinforcement Learning

Anonymous submission

S1. Scope and Evaluation Context

These additional plots accompany the study of *signed critic errors* in Deep Deterministic Policy Gradient (DDPG), Twin Delayed Deep Deterministic Policy Gradient (TD3), and Soft Actor-Critic (SAC) on Pendulum-v1. They illustrate (i) the relationship between predicted critic values and Monte Carlo (MC) return references, (ii) the sign and evolution of critic–reference discrepancies, and (iii) the apparent reduction in those discrepancies after Gaussian-process-regression (GPR) correction. They are descriptive diagnostics, not evidence that the corrected critic has been deployed to improve a learned policy.

As described in the accompanying manuscript, evaluation uses five random seeds (0–4), with 50 recorded checkpoints per seed between episodes 200 and 10,000. For each checkpoint, the procedure samples 50 state–action observations and constructs rollout-based reference estimates using 10 MC rollouts and discount factor $\gamma = 0.99$. Let $q(s, a)$ denote a selected critic estimate and $\widehat{G}(s, a)$ the corresponding rollout reference. Signed discrepancy and the proposed correction take the form

$$e(s, a) = q(s, a) - \widehat{G}(s, a), \qquad Q_c(s, a) = q(s, a) - \lambda(s, a)\mu_e(s, a), \tag{1}$$

where μ_e is the GPR-predicted residual and λ attenuates corrections when predictive uncertainty is high. Positive e represents a critic estimate above the reference, while negative e represents an estimate below it. The diagnostic comparisons presented here are drawn from the uploaded plots; the curves alone do not establish unbiased or calibrated value estimation.

S2. Additional Diagnostic Plots

Critic estimates versus rollout returns. Figure 1 compares the critic-based estimate (red) with the MC return reference (blue) through training. For DDPG, the reported critic estimate is substantially above the reference early in training and the gap narrows later. For TD3, the critic estimate generally lies below the reference after the initial rise; for SAC the two trajectories become closer toward the later checkpoints. These different patterns motivate modeling *signed* errors rather than considering overestimation alone. Because the algorithms optimize different critic targets, particularly the entropy-regularized SAC objective, a reward-only MC return should not be interpreted as a perfectly matched target for all three methods.

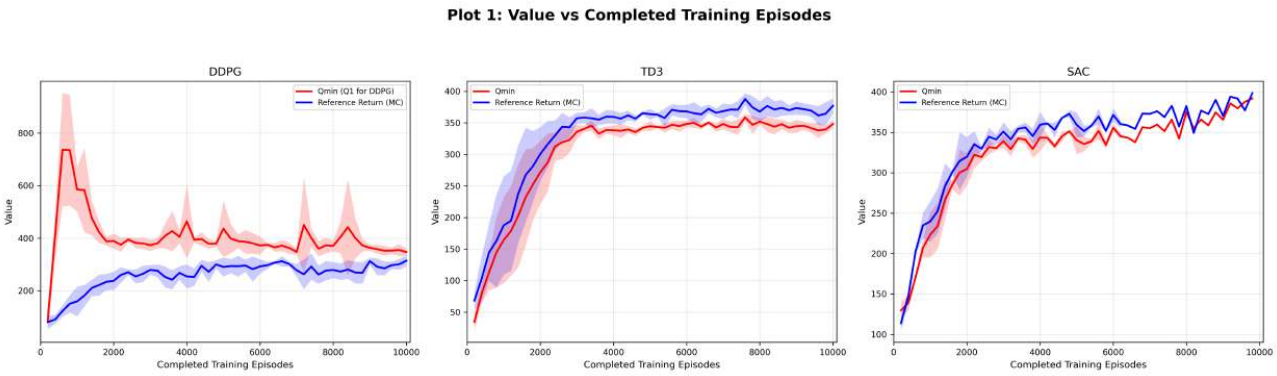


Figure 1: Uploaded diagnostic of critic-based values (red) and Monte Carlo reference returns (blue) versus completed training episodes for DDPG, TD3, and SAC on Pendulum-v1. Shaded regions are present in the source plot; their precise statistical definition should be verified from the original plotting code.

Direction of uncorrected critic discrepancies. Figure 2 makes the sign of the discrepancy explicit. The DDPG trace is predominantly positive, indicating estimates above the displayed reference, whereas TD3 is predominantly negative and SAC is closer to zero with predominantly negative periods. The trace magnitudes change substantially over training; thus an aggregate signed average can obscure large transient errors. This plot illustrates direction and temporal variation, but is not an independent evaluation of the proposed correction.

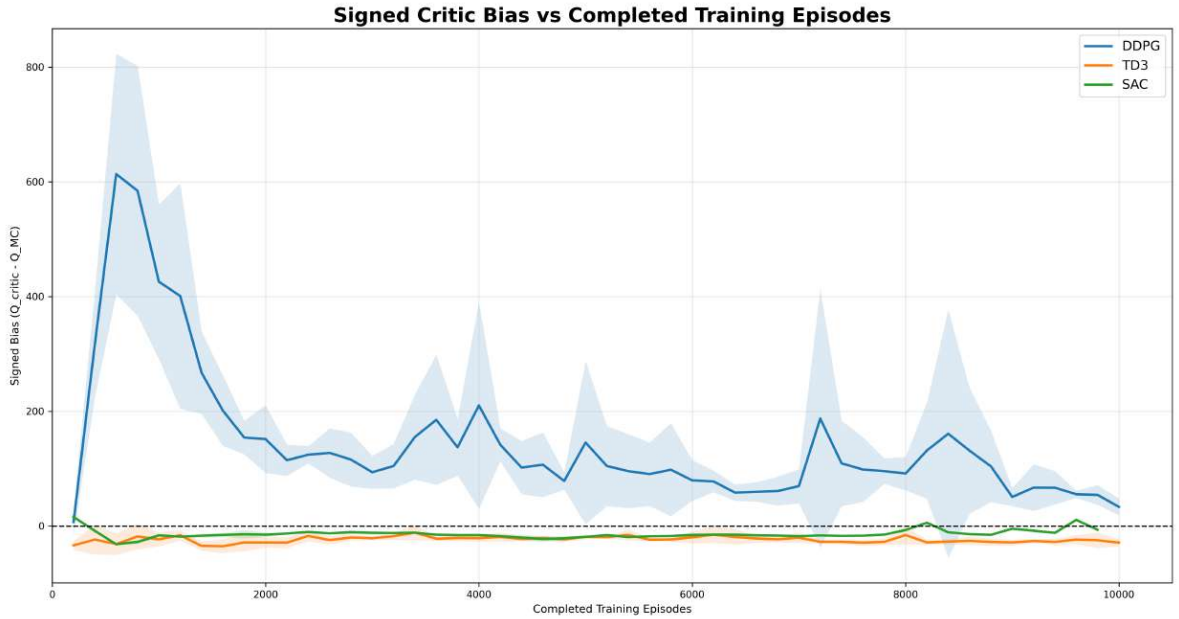


Figure 2: Uploaded signed critic–reference bias traces for DDPG, TD3, and SAC versus completed training episodes. The zero line separates overestimation relative to the selected reference from underestimation. The plotted scale should be reconciled with any numerical summary before reporting equivalent metrics.

GPR-corrected fitting-sample discrepancies. Figure 3 displays raw signed discrepancies (dashed red) together with GPR-corrected discrepancies (green) and the zero-reference line (blue). The corrected traces lie near zero on the displayed training/evaluation points, especially relative to the larger raw deviations shown for DDPG. This is consistent with the GPR residual model fitting the observed discrepancy structure. However, the GPR model is fitted and evaluated on the same state–action observations in the reported diagnostic; consequently, these near-zero traces cannot establish out-of-sample correction quality or improved policy returns.

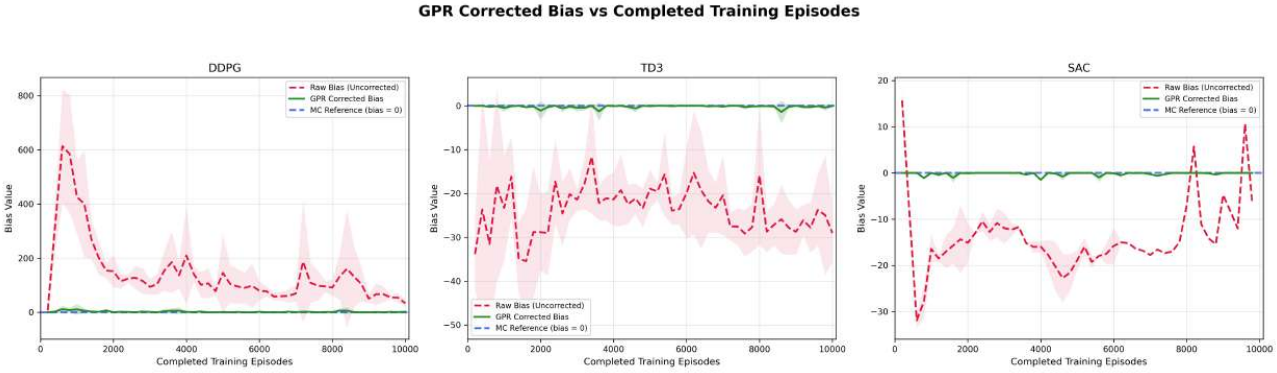


Figure 3: Uploaded raw (red dashed) and GPR-corrected (green) critic–reference discrepancies for DDPG, TD3, and SAC, with zero-reference lines (blue). The corrected curves are fitting-sample diagnostics, not held-out evaluation results.

S3. Interpretation, Consistency Checks, and Limitations

These plots support the *motivation* for a probabilistic residual learner: different critic architectures may exhibit discrepancies of different signs, and a regression model can reproduce local residual patterns on sampled observations. They do *not* yet demonstrate that residual predictions generalize to unseen state–action pairs, that corrected estimates improve policy optimization, or that uncertainty-based correction strength is calibrated.

Before interpreting these graphics quantitatively alongside the main manuscript, the underlying aggregation and units require verification. In particular, the signed-bias plot displays values with magnitudes that differ markedly from the main paper's table of absolute checkpoint-mean discrepancies. Those figures should not be treated as the same metric without a documented explanation of averaging, scaling, or preprocessing differences.

Four further limitations are material. First, evaluation on the fitting data can make the corrected residual appear artificially small; a disjoint training/evaluation split is required. Second, the manuscript notes a mismatch between Pendulum environment-action scales and the action representation used by some critics; this should be

corrected before drawing numerical conclusions. Third, truncated reward-only MC estimates may not match the critics' objectives, especially for SAC's entropy-regularized critic. Fourth, GPR failure-handling paths that substitute the observed error as a correction can produce spuriously zero corrected discrepancies and should be logged separately.

S4. Future Evaluation Protocol

To test the broader Blue Sky proposition, subsequent studies should train the residual model on one set of state–action observations and evaluate it on held-out observations and separate seeds, using action encodings and return targets consistent with each critic. The evaluation should report signed error, absolute error, calibration of predictive uncertainty, fitting failures, and the cost of GPR fitting. Most importantly, an integrated actor–critic experiment should compare uncorrected critics, fixed corrections, and uncertainty-weighted corrections in terms of actual policy returns and training stability. Such a study could determine when learning *how a critic is wrong* provides useful information for the agent's future decisions.